

Sampling Round Complexity of Diffusion Language Models

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1. Introduction and Literature Review

Diffusion language models (DLMs) generate sequences through repeated parallel denoising. Starting from a masked sequence, a predictor reads the current state and returns a distribution for each token position. An update policy then selects positions, whose new values are sampled conditionally independently in parallel. I call one such parallel update a **round**. The central complexity question is how many rounds are necessary to generate a prescribed target distribution.

To make this round measure meaningful, the computation performed inside one round must be restricted. This project assumes that each predictor and update policy is represented by a polynomial-size AC^0 circuit, namely a constant-depth AND/OR/NOT circuit with unbounded fan-in. This is motivated by the fact that constant-precision, constant-depth Transformers of polynomial width can be simulated by AC^0 circuits [4]. Under this assumption, a D -round DLM can be unrolled into a randomized circuit of depth $O(D)$. This one-way abstraction neither covers all practical Transformers nor turns every AC^0 construction into a Transformer.

The second modeling choice is whether generated tokens may be revised. In a **no-revision DLM**, a token is immutable once revealed. In a **DLM with revision**, a revealed token may be overwritten or masked again. Revision acts as writable discrete workspace because temporary information can be created and later replaced by final output tokens.

With both the DLM round and its computational restriction in place, existing results can be compared precisely. Jiang, Haghtalab, and Chen [1] connect DLM generation with sampling circuits and show how intermediate workspace allows a depth- d sampling computation to be simulated in d DLM rounds. They also give a parity separation: $O(1)$ rounds with revision versus $\omega(1)$ without revision. Svete and Sabharwal [5] show that a revision-enabled L-uniform deterministic DLM can recognize every regular language in $O(\log n)$ rounds using $O(n)$ padded workspace positions. These works establish useful recognition and sampling capabilities, but they do not tightly characterize the sampling round complexity of general regular-language input-output pairs.

Inspired by Viola's distribution-complexity framework [2], including its study of distributions of the form $(X, b(X))$, I consider the following **input-output-pair sampling problem** in the circuit regime. For a regular language $L \subseteq \{0, 1\}^*$, let $f_L(x) = \mathbf{1}[x \in L]$ and define

$$\mathcal{D}_{L,n}^{\text{pair}} = \text{Unif}\{(x, f_L(x)) : x \in \{0, 1\}^{n-1}\}.$$

The sequence contains only the n target coordinates, with no additional padded workspace positions.

The sampler must jointly generate a uniform prefix x and its correct membership bit. Ordinary hardness of computing $f_L(x)$ does not automatically make this sampling task hard. Viola's later work [3], for example, develops sampling-specific lower-bound techniques and constructs an explicit function h for which small bounded-depth circuits cannot closely sample $(U, h(U))$. His result shows why sampling requires separate arguments, but the hard function does not directly provide a regular-language lower bound for revision-enabled DLMs.

2. Limitations of Existing Work and Research Gap

Existing work already separates revision from no revision for parity, but the known comparison is $O(1)$ versus $\omega(1)$ rather than a tight regular-language characterization. Parity itself cannot provide a growing lower bound with revision because its revision-enabled sampler takes constant rounds. Meanwhile, Transformer and DLM expressivity results with padding or reusable state mainly study recognition or circuit simulation. Classical regular-language circuit lower bounds also concern recognizing a supplied string and cannot be transferred to sampling without an additional reduction. Viola’s 2020 theorem supplies a sampling perspective, but it applies to a different explicit function and does not yet yield the desired regular-language result with revision.

My current work gives a comparatively complete picture **without revision**. The remaining gap is the revision-enabled case. Revision invalidates the irreversible support trace used by the no-revision lower bound and makes some regular languages, including parity, sampleable in constant rounds. The main open question is whether some regular language still requires $\Omega(\log n / \log \log n)$ rounds when revision is allowed but each round remains in AC^0 .

3. Research Approach and Preliminary Results

3.1 Model and circuit correspondence

I use the two update models from my technical note:

No-revision DLM

Initialize $x \leftarrow M^n$

for $t = 1, \dots, D$:

$S \leftarrow F(x)$, where S is a subset of $\{i : x_i = M\}$

 for each i in S independently:

$x_i \sim p_i(\cdot \mid x)$ over $\{0, 1\}$

 all positions outside S stay unchanged

Output x

DLM with revision

Initialize $x \leftarrow M^n$

for $t = 1, \dots, D$:

 for each i in $[n]$ independently:

$x_i \sim p_i(\cdot \mid x)$ over $\{0, 1, M\}$

Output x

For either model, a valid sampler must output a sequence containing no M . The predictor p and, in the no-revision model, policy F use polynomial-size AC^0 circuits. Each token update uses polynomially many fair random bits, so every exactly represented transition probability is dyadic. Unrolling D rounds gives a randomized circuit of size $\text{poly}(n, D)$ and depth $O(D)$.

3.2 Preliminary no-revision result

The preliminary result is a tight $\Theta(\log n / \log \log n)$ bound. The upper bound uses a $\Theta(\log n)$ -ary finite-monoid product tree, with mask patterns carrying temporary block profiles. For the lower bound, the zero-probability support condition converts a D -round sampler into a depth- $O(D)$ randomized recognizer by exploiting the rigidity of the transition probabilities.

3.3 Proposed revision-enabled direction

Because parity already has an $O(1)$ -round sampler with revision, I will investigate other regular languages as candidates for a growing lower bound. I will first test whether circuit-depth arguments suffice. If not, the proof may need DLM-specific restrictions: positions are sampled independently conditioned on the current state, and only the n output positions persist across rounds, with no padded workspace. Viola’s sampling perspective is a possible guide, but adapting it to a regular language with revision remains open.

4. Expected Outcomes, Challenges, and Feasibility

The expected outcomes are a rigorous no-revision $\Theta(\log n / \log \log n)$ result with explicit assumptions and, for revision, either a concrete lower bound or a partial characterization of what it requires. A further goal is to determine which circuit-level results extend to Transformers. Lower bounds may transfer through the Transformer-to- AC^0 simulation [4], whereas upper bounds require an explicit Transformer construction.

The challenges are the gap between recognition and sampling, the loss of support tracing under revision, probability representation, and explicit Transformer realization. The preliminary no-revision arguments make the project feasible; a conditional revision result or a precise obstacle would also be meaningful.

5. Next-Step Plan

Time	Work
Week 5	Finalize definitions, literature review, and no-revision proofs
Weeks 6–8	Study revision and Viola-inspired ideas
Weeks 9–10	Assess Transformer transfer and write the final report

References

- [1] Haozhe Jiang, Nika Haghtalab, and Lijie Chen. *Diffusion Language Models are Provably Optimal Parallel Samplers*. ICLR, 2026.
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